

8/14 What happened?

1. Convert 100° to radians.
 (a) $\frac{\pi}{18}$ (b) $\frac{5\pi}{9}$ (c) $\frac{10\pi}{9}$ (d) $\frac{18000}{\pi}$
2. Convert $\frac{7\pi}{18}$ into degrees.
 (a) 0.007° (b) 70° (c) 140° (d) 219.911°
3. What is the size of the radius with an arc length of 10 cm and a central angle of 100° ?
 (a) $\frac{18}{\pi}$ cm (b) 10 cm (c) $\frac{50\pi}{9}$ cm (d) 1000 cm
4. Which of the following ratios will be positive?
 (a) $\csc\left(\frac{4\pi}{3}\right)$ (b) $\cot\left(\frac{5\pi}{6}\right)$ (c) $\sec\left(\frac{5\pi}{4}\right)$ (d) $\cot\left(\frac{7\pi}{6}\right)$
5. If $\sin \theta = -\frac{1}{2}$, then what could θ be when $0 \leq \theta \leq 2\pi$?
 (a) $\frac{\pi}{6}, \frac{5\pi}{6}$ (b) $\frac{5\pi}{6}, \frac{7\pi}{6}$ (c) $\frac{5\pi}{6}, \frac{11\pi}{6}$ (d) $\frac{7\pi}{6}, \frac{11\pi}{6}$
6. The terminal arm of an angle in standard position passes through the point $(3, -2)$. Find the radian value of the angle to the nearest hundredth.
 (a) 0.59 (b) -3.73 (c) 5.30 (d) 5.70
7. Which of the following is equivalent to $\cos\left(\frac{4\pi}{3}\right)$? $-\frac{1}{2}$
 (a) $-\cos\left(\frac{2\pi}{3}\right)\frac{1}{2}$ (b) $\cos\left(\frac{\pi}{3}\right)\frac{1}{2}$ (c) $-\cos\left(\frac{\pi}{3}\right) - \frac{1}{2}$ (d) $\cos\left(\frac{5\pi}{3}\right)\frac{1}{2}$
8. The minimums of $y = \sin \theta$ occur according to what values of θ (where $\theta \in \mathbb{R}$, $k \in \mathbb{Z}$).
 (a) $\theta = \frac{\pi}{2} + k\pi$ (b) $\theta = \frac{3\pi}{2} + k\pi$ (c) $\theta = \frac{\pi}{2} + 2k\pi$ (d) $\theta = \frac{3\pi}{2} + 2k\pi$
9. What is the period of $f(x) = -2 \cos\left(\frac{\theta}{3} - \frac{\pi}{2}\right) + 5$?
 (a) $\frac{1}{3}$ (b) 6π (c) $\frac{2\pi}{3}$ (d) 3
10. For the same function as in question 9, what is the maximum y-value?
 (a) -7 (b) -3 (c) 3 (d) 7